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1. 20 points) Consider the toy data set $\{([0, 0], -1), ([2, 2], -1), ([2, 0], +1)\}$. Set up the dual problem for the toy data set. Then, solve the dual problem and compute $\alpha^*$, the optimal Lagrange multipliers. (Note that there will be three weights $w = [w0, w1, w2]$ by considering the bias.)

   Dataset: $\{([0, 0], -1), ([2, 2], -1), ([2, 0], +1)\}$

   Need three weights $w = [w0, w1, w2]$ and only two data coordinates, so just add a 1 to the vectors as bias.

   x Vectors:

   $$x1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \rightarrow y1 = -1$$

   $$x2 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \rightarrow y2 = -1$$

   $$x3 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \rightarrow y3 = 1$$

   Classification equation is $f(x) = w0 + w1x1 + w2x2$

   $L(w, \alpha) = \frac{1}{2}||w||^2 - \sum(i = 1, 3)\alpha_i(y_i w^T x_i - 1)$

   SVM dual derivation, since we have the $\alpha_i$ values, can recover $w$

   Take the gradient with respect to $w$ and set to 0:

   $\nabla_w L = w - \sum(i = 1, 3)\alpha_i y_i x_i = 0$

   $w = \sum(i = 3, w)\alpha_i y_i x_i$

   Substitute $w$ into the L equation to then get dual problem:

   $\max(\alpha \geq 0)\ \sum(i = 1, 3)\alpha_i - \frac{1}{2} * \sum(i = 1, 3)\sum(j = 1, 3)\alpha_i \alpha_j \alpha y_i \alpha_j (x_i^T x_j)$

   Then get inner product $x_i^T x_j$

   $$K = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 9 & 5 \\ 1 & 5 & 5 \end{bmatrix}$$

   $Q_{ij} = y_i y_j K_{ij}$

   $$Q = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 9 & -5 \\ -1 & -5 & 5 \end{bmatrix}$$

   Therefore dual objective is $W(\alpha) = \alpha_1 + \alpha_2 + \alpha_3 - \frac{1}{2}\alpha^T Q \alpha$, where $\alpha_i \geq 0$

   Since $\nabla_\alpha W = 0$, $1 - Qa - 0 \rightarrow Qa = 1$

   $$Q = \begin{bmatrix} 1 & 1 & -1 \\ 1 & 9 & -5 \\ -1 & -5 & 5 \end{bmatrix}$$

   $\alpha = [\alpha_1, \alpha_2, \alpha_3]$

Row 1: $\alpha_1 + \alpha_2 - \alpha_3 = 1$

Row 2: $\alpha_1 + 9\alpha_2 - 5\alpha_3 = 1$

Row 3: $-\alpha_1 - 5\alpha_2 + 5\alpha_3 = 1$

$\alpha_1 = 1 - \alpha_2 + \alpha_3$

Plug $\alpha_1$ into row 2

$(1 - \alpha_2 + \alpha_3) + 9\alpha_2 - 5\alpha_3 = 1$

$8\alpha_2 - 4\alpha_3 = 0$

$2\alpha_2 - \alpha_3 = 0$

$\alpha_3 = 2\alpha_2$

Then plug $\alpha_1$ into row 3

$-(1 - \alpha_2 + \alpha_3) - 5\alpha_2 + 5\alpha_3 = 1$

$-4\alpha_2 + 4\alpha_3 = 2$

$-2\alpha_2 + 2\alpha_3 = 1$

Then plug $\alpha_3$ into equation

$-2\alpha_2 + 2(2\alpha_2) = 1$

$2\alpha_2 = 1 \rightarrow \alpha_2 = \frac{1}{2}$

Then solve for $\alpha_3$

$\alpha_3 = 2(\alpha_2) = 1$

Then solve for $\alpha_1$

$\alpha_1 = 1 - \alpha_2 + \alpha_3 = 1 - 0.5 + 1 = 1.5$

So, $\alpha^* = [1.5, 0.5, 1]$ and since all $\alpha$ values are greater than 0, this answer works and is the optimal solution

Therefore, $\alpha^* = (\alpha_1, \alpha_2, \alpha_3) = (1.5, 0.5, 1)$

And for finding the three weights:

$w = \sum(i = 1, 3)\alpha_i y_i x_i$

$x1$: $1.5(-1)\begin{bmatrix}1\\0\\0\end{bmatrix} = \begin{bmatrix}-1.5\\0\\0\end{bmatrix}$

$x2$: $0.5(-1)\begin{bmatrix}1\\2\\2\end{bmatrix} = \begin{bmatrix}-0.5\\-1\\-1\end{bmatrix}$

$x3$: $1.0(1)\begin{bmatrix}1\\2\\0\end{bmatrix} = \begin{bmatrix}1\\2\\0\end{bmatrix}$

$w1 = -1.5 - 0.5 + 1 = -1$

$w2 = 0 - 1 + 2 = 1$

$w3 = 0 - 1 + 0 = -1$

Therefore, $w^* = [w0, w1, w2] = [-1, 1, -1]$

2. (20 points) In a separable case, when a multiplier $\alpha_i > 0$, its corresponding data point $(x_i, y_i)$ is on the boundary of the optimal separating hyperplane with $y_i(w^T x_i) = 1$. Show that the inverse is not True. Namely, it is possible that $\alpha_i = 0$ and $(x_i, y_i)$ is on the boundary satisfying $y_i(w^T x_i) = 1$.

[Hint: Consider a toy data set with two positive examples at ([0,0], +1) and ([1, 0], +1), and one negative example at ([0, 1], -1).] (Note that there will be three weights $w = [w0, w1, w2]$ by considering the bias.)

For hard-margin SVM, each constraint is $y_i(w^T x_i) \geq 1$

KKT complementary slackness says $\alpha_i(y_i(w^T x_i) - 1) = 0$, $\alpha_i \geq 0$

If $\alpha_i > 0$, only way product can be 0 is $y_i(w^T x_i) - 1 = 0 \rightarrow y_i(w^T x_i) = 1$

- This means the point is on the boundary

Also $y_i(w^T x_i) - 1 = 0$ is true, which means $\alpha_i * 0 = 0$ when plugged into the slack equation, and shows how $\alpha_i$ can be 0 or positive. Dataset can help prove the inverse.

Dataset:

([0, 0], 1)

([1, 0], 1)

([0, 1], -1)

Bias is a weight, so represent it as a 1 in the vector $x = [1, x1, x2]$ and $w = [w0, w1, w2]$

$x1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ and $y1 = 1$

$x2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $y2 = 1$

$x3 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $y3 = -1$

$y_i(w^T x_i) \geq 1$ is constraint

Point 1: $w^T x1 = w0 \geq 1$

Point 2: $w^T x2 = w0 + w1 \geq 1$

Point 3: $y3(w^T x3) = -1(w0 + w2) \geq 1 \rightarrow w0 + w2 \leq -1$

Pick smallest values that satisfy the constraint to minmize the $\frac{1}{2}||w||^2$

For point 1, that would be $w1 = 1$

For point 3, plug in $x1$ and it is $1 + w2 \leq -1$, which means $w2 = -2$

For point 2, plug in $x1$ and it is $1 + w1 \geq 1 \rightarrow w1 \geq 0$, which means $w1 = 0$

Therefore, $w^* = [1, 0, -2]$ and all of the points are on margin

Checking if the points fall on the boundary:

Point 1: $y1(w^T x1) = 1 * 1 = 1$

Point 2: $y2(w^T x2) = 1 * (1 + 0) = 1$

Point 3: $y3(w^Tx3) = -1 * (1-2) = (-1) * (-1) = 1$

All 3 points are on the boundary, since the $y_n(w^Tx_n)$ for all 3 $n$ values equate to 1, which is the decision boundary line.

Then have information needed to solve dual problem

$Q_{ij} = y_iy_j(x_i^Tx_j)$ and $W(\alpha) = 1^T\alpha - 1/2\alpha^TQ\alpha$

Then need to compute the inner products $(x_i^Tx_j)$

$$x1 * x1 = \begin{bmatrix}1\\0\\0\end{bmatrix} * \begin{bmatrix}1\\0\\0\end{bmatrix} = 1$$

$$x1 * x2 = \begin{bmatrix}1\\0\\0\end{bmatrix} * \begin{bmatrix}1\\1\\0\end{bmatrix} = 1$$

$$x1 * x3 = \begin{bmatrix}1\\0\\0\end{bmatrix} * \begin{bmatrix}1\\0\\1\end{bmatrix} = 1$$

$$x2 * x2 = \begin{bmatrix}1\\1\\0\end{bmatrix} * \begin{bmatrix}1\\1\\0\end{bmatrix} = 2$$

$$x2 * x3 = \begin{bmatrix}1\\1\\0\end{bmatrix} * \begin{bmatrix}1\\0\\1\end{bmatrix} = 1$$

$$x3 * x3 = \begin{bmatrix}1\\0\\1\end{bmatrix} * \begin{bmatrix}1\\0\\1\end{bmatrix} = 2$$

So $K = \begin{bmatrix}1 & 1 & 1\\1 & 2 & 1\\1 & 1 & 2\end{bmatrix}$

Labels $y = [1, 1, -1]$

And $Q_{ij} = y_iy_jK_{ij}$

$Q11 = (1)(1) * 1 = 1$

$Q12 = (1)(1) * 1 = 1$

$Q13 = (1)(-1) * 1 = -1$

$Q21 = (1)(1) * 1 = 1$

$Q22 = (1)(1) * 2 = 2$

$Q23 = (1)(-1) * 1 = -1$

$Q31 = (-1)(1) * 1 = -1$

$Q32 = (-1)(1) * 1 = -1$

$Q33 = (-1)(-1) * 2 = 2$

So $Q = \begin{bmatrix}1 & 1 & -1\\1 & 2 & -1\\-1 & -1 & 2\end{bmatrix}$

$\nabla W(\alpha) = 0 \rightarrow 1 - Q\alpha = 0 \rightarrow Q\alpha = 1$

Let $\alpha = \begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}$

$\alpha_1 + \alpha_2 - \alpha_3 = 1$

$\alpha_1 + 2\alpha_2 - \alpha_3 = 1$

$-\alpha_1 - \alpha_2 + 2\alpha_3 = 1$

$(\alpha_1 + 2\alpha_2 - \alpha_3) - (\alpha_1 + \alpha_2 - \alpha_3) = 1 - 1$

$\alpha_2 = 0$

Then plug $\alpha_2$ into first equation

$\alpha_1 + 0 - \alpha_3 = 1$

$\alpha_1 = 1 + \alpha_3$

Then plug $\alpha_2$ and $\alpha_1$ into the third equation

$-(1 + \alpha_3) + 2\alpha_3 = 1$

$-1 - \alpha_3 + 2\alpha_3 = 1$

$-1 + \alpha_3 = 1$

$\alpha_3 = 2$

And $\alpha_1 = \alpha_3 + 1$ so $\alpha_1 = 2 + 1 = 3$

$\alpha_1 = 3$, $\alpha_2 = 0$, $\alpha_3 = 2$

Therefore $\alpha^* = (3, 0, 2)$

It was proven before that $y2(w^T x2) = 1$ and therefore point 2 is on the boundary line in the dataset. Now point 2's multiplier has been proven as $\alpha_2 = 0$. Therefore it is possible for $\alpha_i = 0$ and $y_i(w^T x_i)$ and that the inverse statement in the problem is not true.

3. 3. Kernel Function: A function K computes $K(x_i, x_j) = -x_i^T x_j$. Is this function a valid kernel function for SVM? Prove or disprove it.

   A function K is a kernel iff for any finite set of points, the corresponding kernel matrix is symmetric and positive semi-definite.

   - So need to prove or disprove using the positive semi-definite condition.

   Use dataset $x1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $x2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and plug into $K(x_i, x_j) = -x_i^T x_j$

   $K(x1, x1) = -(x1^T x1) = -(1) = -1$

   $K(x1, x2) = -(x1^T x2) = -(0) = 0$

   $K(x2, x1) = -(x2^T x1) = -(0) = 0$

   $K(x1, x1) = -(x2^T x2) = -(1) = -1$

   Kernel matrix is then $G = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

   Positive semi-definite condition means that for $\forall c \in \mathbb{R}^2$, $c^T G c \geq 0$

   Choose the first point $c = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ so that

$c^TGc = \begin{bmatrix}1\\0\end{bmatrix}^T \begin{bmatrix}-1 & 0\\0 & -1\end{bmatrix} \begin{bmatrix}1\\0\end{bmatrix}$

$Gc = \begin{bmatrix}-1 & 0\\0 & -1\end{bmatrix} \begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}(-1*1+0*0)\\(0*1+-1*0)\end{bmatrix} = \begin{bmatrix}(-1)\\(0)\end{bmatrix}$

$\begin{bmatrix}(-1)\\(0)\end{bmatrix} * \begin{bmatrix}1 & 0\end{bmatrix} = 1*-1+0*0 = -1$

So $c^TGc = -1$ and is not positive. This violates the positive semi-definite condition of a prototypical kernel function. Therefore it is proven through counterexample that the function $K(x_i, x_j) = -x_i^T x_j$ is not a valid SVM kernel function.

$\alpha$ $\nabla$ $\sum$